

Entity Relationship Model / Domain Data Model

Institute Management System (IMS)

Version: 1.1

Based On: DDD Context Map v1.2

Scope: Single-client implementation

Excluded: Tenant Setup, SaaS Subscription Management, CMS / Website Builder

Architecture Direction: Next.js Monorepo Modular Architecture

1. Purpose

This document defines the initial domain data model for the IMS platform. It identifies the core entities, relationships, ownership boundaries, and important data rules needed for database design, API planning, and UI development.

This is not yet a physical database schema. It is a business-level entity model that can later be converted into tables, Prisma models, SQL schema, or API contracts.

2. Effective Dating & Status Management

Many master and configuration entities require lifecycle management.

To support future business requirements, the following fields should be added to entities where records may become active, inactive, deprecated, replaced, or valid only during a specific period.

Status

EffectiveStartDate

EffectiveEndDate

Purpose

Field	Purpose
Status	Controls whether the record is Active, Inactive, Draft, Archived, etc.
EffectiveStartDate	Date from which the record becomes valid
EffectiveEndDate	Date until which the record remains valid
NULL EffectiveEndDate	Indicates the record remains valid indefinitely

Examples

Course

A course may only be offered during a specific period.

```
Course A
EffectiveStartDate = 2025-01-01
EffectiveEndDate = 2025-12-31
```

Course Pricing

Pricing may change annually.

```
Course Price 2025
EffectiveStartDate = 2025-01-01
EffectiveEndDate = 2025-12-31

Course Price 2026
EffectiveStartDate = 2026-01-01
EffectiveEndDate = NULL
```

Certificate Template

A new certificate design may replace an old one.

```
Template V1
EffectiveEndDate = 2025-06-30

Template V2
EffectiveStartDate = 2025-07-01
```

Recommended Rule

Any entity that represents:

- Master Data
- Configuration Data
- Pricing Data
- Rules
- Templates
- Organizational Structures
- Contracts
- Assignments

should include:

Status
EffectiveStartDate
EffectiveEndDate

unless the entity is purely transactional.

3. Core Data Model Overview

The central concept of the IMS system is **Enrollment**.

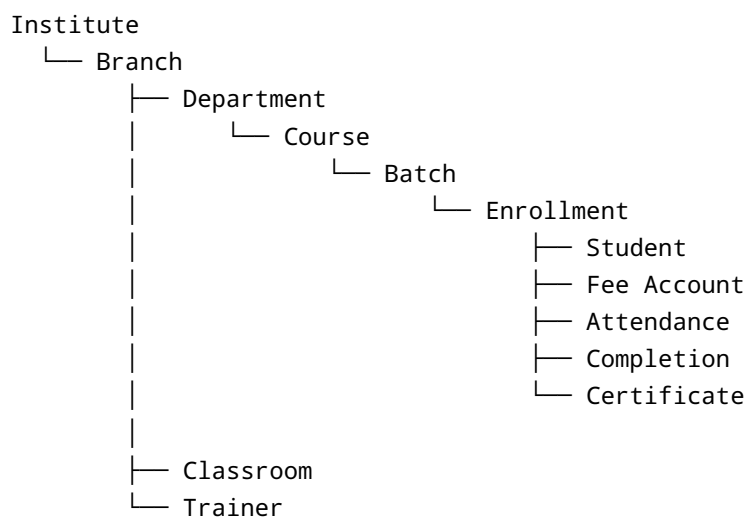
Enrollment connects:

- Student
- Course
- Batch
- Branch
- Fee
- Attendance
- Completion
- Certificate

Enrollment supports three learning flows:

Regular Enrollment
Corporate Enrollment
Walk-In Enrollment

4. High-Level Entity Relationship Map



```
Lead
├─ Admission
│   └─ Student
│       └─ Enrollment

Corporate Account
├─ Corporate Contract
│   └─ Corporate Program
│       └─ Corporate Participant
│           └─ Enrollment

Trainer
├─ Trainer Assignment
│   └─ Batch / Session

Finance
├─ Fee Plan
├─ Payment
├─ Receipt
├─ Discount
└─ Refund

Document
├─ Student Document
└─ Trainer Document

Audit Log
├─ Tracks critical actions across all modules
```

5. Foundation Entities

5.1 Institute

Represents the client institute.

Key Fields

```
InstituteId
InstituteName
LegalName
PrimaryEmail
PrimaryPhone
Address
Country
DefaultCurrency
DefaultLanguage
```

Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt

Relationships

Relationship	Type
Institute has many Branches	1
Institute has many Users	1

5.2 Branch

Key Fields

BranchId
InstituteId
BranchName
BranchCode
Address
City
Country
Phone
Email
Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt

5.3 Department

Key Fields

DepartmentId
BranchId
DepartmentName
DepartmentCode
Description
Status
EffectiveStartDate
EffectiveEndDate

CreatedAt
UpdatedAt

5.4 Classroom

Key Fields

ClassroomId
BranchId
ClassroomName
Capacity
Location
Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt

6. Identity & Access Entities

6.1 User

Key Fields

UserId
BranchId
FullName
Email
PasswordHash
Phone
UserType
Status
EffectiveStartDate
EffectiveEndDate
LastLoginAt
CreatedAt
UpdatedAt

6.2 Role

```
RoleId
RoleName
Description
Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt
```

6.3 Permission

```
PermissionId
PermissionCode
PermissionName
ModuleName
ActionName
Description
Status
EffectiveStartDate
EffectiveEndDate
```

6.4 UserRole

```
UserRoleId
UserId
RoleId
AssignedAt
AssignedBy
Status
EffectiveStartDate
EffectiveEndDate
```

6.5 RolePermission

```
RolePermissionId
RoleId
PermissionId
Status
EffectiveStartDate
```

```
EffectiveEndDate  
CreatedAt
```

7. Lead & Inquiry Entities

7.1 Lead

```
LeadId  
LeadNumber  
BranchId  
AssignedCounselorId  
FullName  
Phone  
Email  
LeadSourceId  
LeadStageId  
InterestedCourseId  
LeadType  
Priority  
Status  
CreatedAt  
UpdatedAt
```

Lead is transactional and therefore does not require effective dating.

7.2 LeadSource

```
LeadSourceId  
SourceName  
Description  
Status  
EffectiveStartDate  
EffectiveEndDate
```

7.3 LeadStage

```
LeadStageId  
StageName  
DisplayOrder  
IsWonStage  
IsLostStage
```



```
Status
EffectiveStartDate
EffectiveEndDate
```

7.4 FollowUp

Transactional entity.

```
FollowUpId
LeadId
CounselorId
FollowUpDate
FollowUpType
Notes
Status
CompletedAt
CreatedAt
```

7.5 Campaign

```
CampaignId
CampaignName
CampaignType
StartDate
EndDate
Budget
Description
Status
EffectiveStartDate
EffectiveEndDate
```

8. Admission, Student & Enrollment Entities

8.1 Admission

Transactional entity.

```
AdmissionId
LeadId
StudentId
AdmissionNumber
```

```
AdmissionDate
AdmissionStatus
CreatedBy
CreatedAt
UpdatedAt
```

8.2 Student

```
StudentId
StudentNumber
FullName
Gender
DateOfBirth
Email
Phone
AlternatePhone
Address
Nationality
Status
CreatedAt
UpdatedAt
```

8.3 StudentIdentityField

```
StudentIdentityFieldId
FieldName
FieldCode
FieldType
IsRequired
DisplayOrder
Status
EffectiveStartDate
EffectiveEndDate
```

8.4 StudentIdentityValue

Transactional entity.

```
StudentIdentityValueId
StudentId
StudentIdentityFieldId
FieldValue
```

```
CreatedAt
UpdatedAt
```

8.5 StudentIDCard

```
StudentIDCardId
StudentId
CardNumber
IssuedDate
ExpiryDate
CardStatus
PrintedAt
CreatedAt
```

8.6 Enrollment

Transactional aggregate root.

```
EnrollmentId
EnrollmentNumber
StudentId
CorporateParticipantId
BranchId
CourseId
BatchId
EnrollmentType
EnrollmentDate
EnrollmentStatus
CompletionStatus
CertificateEligibilityStatus
CreatedAt
UpdatedAt
```

9. Corporate Training Entities

9.1 CorporateAccount

```
CorporateAccountId
CompanyName
RegistrationNumber
Industry
```

Phone
Email
Address
Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt

9.2 CorporateContactPerson

CorporateContactPersonId
CorporateAccountId
FullName
Designation
Email
Phone
IsPrimary
Status
EffectiveStartDate
EffectiveEndDate

9.3 CorporateContract

CorporateContractId
CorporateAccountId
ContractNumber
ContractValue
BillingModel
StartDate
EndDate
RenewalDate
SLA
Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt

9.4 CorporateProgram

```
CorporateProgramId  
CorporateContractId  
CourseId  
BranchId  
ProgramName  
StartDate  
EndDate  
Status  
EffectiveStartDate  
EffectiveEndDate
```

9.5 CorporateParticipant

```
CorporateParticipantId  
CorporateAccountId  
CorporateProgramId  
LinkedStudentId  
EmployeeName  
EmployeeCode  
Email  
Phone  
Designation  
Department  
Status  
CreatedAt  
UpdatedAt
```

10. Course & Batch Entities

10.1 Course

Key Fields

```
CourseId  
DepartmentId  
CourseCode  
CourseName  
Description  
CourseType  
DurationType  
DurationValue
```

```
AllowsWalkInCompletion
Status
EffectiveStartDate
EffectiveEndDate
CreatedAt
UpdatedAt
```

Business Rule

A course can only be offered when:

```
CurrentDate >= EffectiveStartDate
AND
CurrentDate <= EffectiveEndDate
```

or

```
EffectiveEndDate IS NULL
```

10.2 CoursePricing

```
CoursePricingId
CourseId
BranchId
CustomerType
BatchType
Currency
BaseAmount
TaxApplicable
Status
EffectiveStartDate
EffectiveEndDate
```

10.3 CourseCompletionRule

```
CourseCompletionRuleId
CourseId
CompletionType
MinimumAttendancePercentage
ExamRequired
ManualApprovalRequired
PaymentRequiredForCertificate
Status
```

```
EffectiveStartDate  
EffectiveEndDate
```

10.4 Batch

```
BatchId  
CourseId  
BranchId  
BatchCode  
BatchName  
StartDate  
EndDate  
Capacity  
BatchStatus  
CreatedAt  
UpdatedAt
```

Batch dates already define validity and therefore separate effective dates are optional.

10.5 BatchTrainer

```
BatchTrainerId  
BatchId  
TrainerId  
Role  
AssignedFrom  
AssignedTo  
Status
```

10.6 WaitingList

```
WaitingListId  
CourseId  
BatchId  
LeadId  
StudentId  
RequestedDate  
Status  
CreatedAt
```

11. Scheduling Entities

11.1 ScheduleSession

```
ScheduleSessionId  
BatchId  
CourseId  
BranchId  
TrainerId  
ClassroomId  
SessionDate  
StartTime  
EndTime  
SessionStatus  
CreatedAt  
UpdatedAt
```

11.2 TrainerAvailability

```
TrainerAvailabilityId  
TrainerId  
AvailableDate  
StartTime  
EndTime  
Status
```

12. Attendance Entities

Attendance entities are transactional and do not require effective dating.

13. Fee & Finance Entities

13.1 FeePlan

```
FeePlanId  
CourseId  
BranchId  
PlanName  
TotalAmount  
Currency
```


TaxApplicable
Status
EffectiveStartDate
EffectiveEndDate

13.2 InstallmentPlan

InstallmentPlanId
FeePlanId
InstallmentName
DueDateOffset
Amount
DisplayOrder
Status
EffectiveStartDate
EffectiveEndDate

13.3 EnrollmentFeeAccount

Transactional entity.

13.4 Payment

Transactional entity.

13.5 Receipt

Transactional entity.

13.6 Discount

Transactional entity.

13.7 Refund

Transactional entity.

14. Faculty / Trainer Entities

14.1 Trainer

```
TrainerId  
BranchId  
FullName  
Email  
Phone  
TrainerType  
Status  
EffectiveStartDate  
EffectiveEndDate  
CreatedAt  
UpdatedAt
```

14.2 TrainerQualification

```
TrainerQualificationId  
TrainerId  
QualificationName  
Institution  
YearCompleted  
CertificateDocumentId  
Status  
EffectiveStartDate  
EffectiveEndDate
```

14.3 TrainerDocument

```
TrainerDocumentId  
TrainerId  
DocumentId  
DocumentType  
Status  
EffectiveStartDate  
EffectiveEndDate
```

14.4 TrainerAssignment

```
TrainerAssignmentId
TrainerId
BatchId
ScheduleSessionId
AssignmentType
StartDate
EndDate
Status
```

14.5 TrainerPayment

Transactional entity.

15. Exam, Result & Completion Entities

15.1 Exam

```
ExamId
CourseId
BatchId
ExamName
ExamDate
MaximumMarks
PassingMarks
Status
```

15.2 Result

Transactional entity.

15.3 CourseCompletion

Transactional entity.

16. Certificate Entities

16.1 CertificateTemplate

```
CertificateTemplateId  
CourseId  
TemplateName  
Language  
TemplateFileUrl  
Status  
EffectiveStartDate  
EffectiveEndDate  
CreatedAt
```

16.2 Certificate

Transactional entity.

16.3 CertificateVerificationLog

Transactional entity.

17. Document Management Entities

17.1 DocumentType

```
DocumentTypeId  
DocumentName  
DocumentCode  
OwnerType  
IsRequired  
Status  
EffectiveStartDate  
EffectiveEndDate
```

17.2 Document

Transactional entity.

17.3 DocumentVerification

Transactional entity.

18. Communication Entities

18.1 CommunicationTemplate

```
CommunicationTemplateId  
TemplateName  
Channel  
Subject  
Body  
Language  
Status  
EffectiveStartDate  
EffectiveEndDate
```

18.2 CommunicationLog

Transactional entity.

18.3 SystemNotification

Transactional entity.

19. Reporting Entities

19.1 ReportDefinition

```
ReportDefinitionId  
ReportName  
ReportCode  
ModuleName  
Description  
Status  
EffectiveStartDate  
EffectiveEndDate
```

19.2 DashboardWidget

```
DashboardWidgetId  
WidgetName  
WidgetCode  
DataSource  
DisplayOrder  
Status  
EffectiveStartDate  
EffectiveEndDate
```

19.3 MetricSnapshot

Transactional snapshot entity.

20. Audit & Compliance Entities

Audit entities are transactional and historical by nature and therefore do not require effective dating.

21. Key Cardinality Summary

No change from the original cardinality model.

22. Common Fields for Most Entities

Most transactional entities should include:

```
CreatedAt  
CreatedBy  
UpdatedAt  
UpdatedBy  
DeletedAt  
DeletedBy  
IsDeleted  
Status
```

Most master/configuration entities should additionally include:

EffectiveStartDate
EffectiveEndDate

Recommended status values:

Draft
Active
Inactive
Suspended
Archived
Expired

For audit-sensitive entities, include:

Version
ChangeReason

23. Recommended Next Document

The next recommended document is:

API Contract Planning / Module-wise API Inventory

This should define:

- API groups
- Endpoint names
- Request payloads
- Response payloads
- Permission requirements
- Validation rules
- Error scenarios

Recommended order:

1. Identity & Access APIs
2. Organization APIs
3. Lead APIs
4. Student & Enrollment APIs
5. Course & Batch APIs
6. Finance APIs
7. Attendance APIs
8. Certificate APIs